

Low-Ropes Force Map Relay

Predict balance and stability before the balance challenge proves you right.

Win condition: Most accurate predictions and clear debrief connections.

THE SCIENCE YOU ARE USING

Center of mass. Your center of mass is the average position of your weight. You stay balanced while it stays over your base of support. Lean too far and the line of gravity leaves the base, so you tip. Lowering your center of mass adds stability.

Mapping forces. At each balance challenge, you predict where balance will be hard, then test it with your body and a partner. The four challenges: (1) stand heels and back against a wall and try to lift one foot forward or reach a coin on the floor ahead of you; (2) sit in a backless chair, torso straight, and try to stand without leaning forward; (3) bend, grab your toes, and try to hop forward; (4) walk a taped floor line holding a small weight at arm's length, then at your chest. Each one fails or works for a center-of-mass reason. Accurate predictions and a clear debrief score.

HOW TO RUN IT

- 01 **Read the challenge.** Look at the next balance challenge card and the body position it asks for.
- 02 **Predict balance points.** On the card, mark whether it will work or fail, and where your center of mass moves.
- 03 **Test it.** Do the challenge near the wall so you can catch yourself. Notice where you actually felt unstable.
- 04 **Map the forces.** Sketch where your weight acted and whether your center of mass stayed over your base of support.
- 05 **Debrief the connection.** Explain how lowering or shifting your center of mass changed stability.

Safety: Do every challenge next to a clear wall so a helper can steady you if needed. Keep bags and unused chairs out of the test area. A helper spots the backless-chair stand. The taped line stays flat on the floor: no running and no raised beam.

MATERIALS

Balance challenge cards	set
Clipboards	6
Center-of-mass templates	per team
Pencils	shared

YOUR PLAN AND DATA

Record one prediction and one observed result for each of the four fixed challenges.

Challenge 1 prediction and observed result: _____

Challenge 2 prediction and observed result: _____

Challenge 3 prediction and observed result: _____

Challenge 4 prediction and observed result: _____

EXPLAIN YOUR FORCE MAP

Which challenge best showed the line of gravity leaving the base of support? _____

How did the center-of-mass position explain one observed result? _____

HOW YOU SCORE (100 POINTS)

SCORING CRITERION	POINTS
Correct predictions	35
Force-map quality	25
Debrief explanation	20
Teamwork reflection	20
Total	100